Markov Information Source and Redundancy

Luping, Xiang (向路平)

luping.xiang@nju.edu.cn

Office: 南雍楼西231

Questions

- If you have one book (novel) written by English, e.g., 100 pages, what is limit of the size for the compressed book if you can apply most effective compression method for storage purpose?

Application of Markov chains

- Markov chains models/methods are useful in answering questions such as:
- How long does it take to shuffle deck of cards?
- How likely is a queue to overflow its buffer?
- How long does it take for a knight making random moves on a chessboard to return to his initial square
- What do the hyperlinks between web pages say about their relative popularities?

Main Content

1. Definition of Markov Source
2. Model of Markov Source
3. Calculation of Entropy Rate
4. Redundancy of Info Source

https://www.bilibili.com/video/BV1HL411t7ce/?spm_id_from=333.337.search-card.all.click

Definition of Markov Source in Textbook

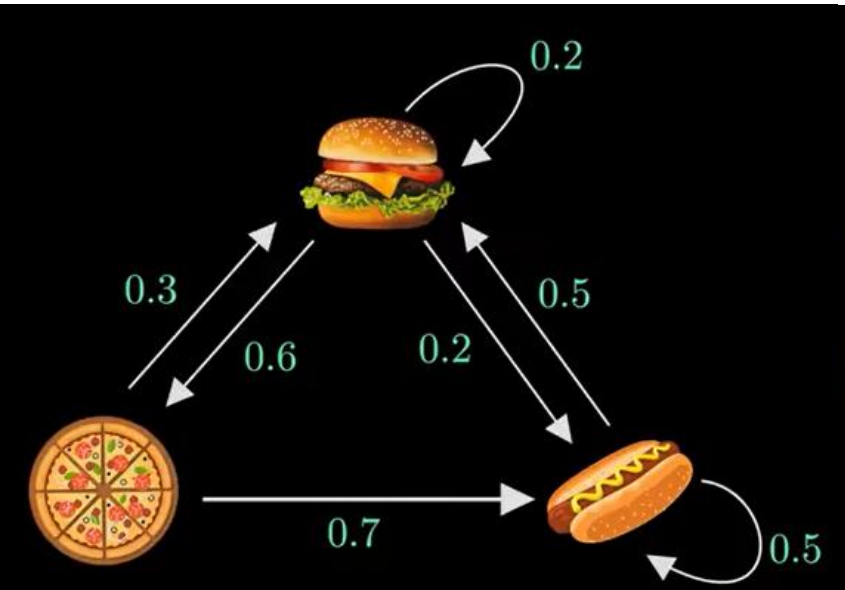
- Such sources are characterized by a set of states, denoted by the integers $1, \ldots, J$, and a source letter alphabet, denoted as usual by $a_1,..,a_K$,
- Each unit of time the source emits a letter and assumes a new state.
- The source sequence is denoted by $\mathrm{u}=(u_1, u_2, \ldots)$ and the state sequence by $\mathrm{s}=(s_1, s_2, \ldots)$
- Let Q_{ij} denote the conditional probability of entering state i given that the previous state was j,

$$Q_{ij}=\Pr(s_l=i \mid s_{l-1}=j) \qquad (3.6.1)$$

- Assume that the probability of entering a state depends only the past state

$$\Pr(s_l \mid s_{l-1}, s_{l-2}, \ldots)= \Pr(s_l \mid s_{l-1}) \qquad (3.6.2)$$

- A random sequence of states obeying (3.6.1) and (3.6.2) is called a finite homogeneous Markov chain.



$$\begin{array}{c} \\ \\ \\ \end{array}\begin{array}{c} \begin{array}{ccc} \text{🍔} & \text{🍕} & \text{🌭} \end{array} \\ \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix} \end{array}$$

$P(X_4 = \text{🌭} \mid X_1 = \text{🍕}, X_2 = \text{🍔}, X_3 = \text{🍕})$

$P(X_4 = \text{🌭} \mid X_3 = \text{🍕}) = 0.7$

https://www.bilibili.com/video/BV1HL411t7ce/?spm_id_from=333.337.search-card.all.click

After ∞ steps...

stationary distribution

$P(\text{🍔})$	$P(\text{🍕})$	$P(\text{🌭})$
0.35191	0.21245	0.43564

Approximation of N-D Info Source

- N-D discrete stationary memory information source is an approximate way to study a practical discrete stationary information source. It cannot accurately express the characteristics of a practical source.
- Suppose that a discrete info source randomly chooses a symbol from the set $\{a_1, a_2, a_3, a_4, a_5\}$ and generates a sequence of the symbols:

Discrete Info Source: $\boldsymbol{X} = a_3,\ a_5, a_1, a_3, a_5, a_2, a_4, a_1, \cdots$

2-D Approximation: $\boldsymbol{X} = a_3,\ a_5, \underbrace{a_1, \boxed{a_3, a_5, a_2}, a_4}_{\textbf{Uncorrelated}}, a_1, \cdots$

3-D Approximation: $\boldsymbol{X} = a_3,\ \underbrace{a_5, \boxed{a_1, a_3, a_5, a_2, a_4}, a_1}_{\textbf{Uncorrelated}}, \cdots$

∞-D Approximation: $\boldsymbol{X} = \boxed{a_3,\ a_5, a_1, a_3, a_5, a_2, a_4, a_1, \cdots}$

All the symbols are correlated

Limited Memory of Info. Source

- The correlation among the symbols transmitted by a information source is limited. At an arbitrary moment, the symbol transmitted is only correlated to limited number of symbols previously transmitted. This is the Markov property.

Discrete Info. Source: $\boldsymbol{X} = a_3,\ a_5, a_1, a_3, a_5, a_2, a_4, a_1, \cdots$

Markov Info. Source:

$\boldsymbol{X} = a_3,\ a_5,$	Moment 2;
$\boldsymbol{X} = a_3,\ a_5, a_1$	Moment 3;
$\boldsymbol{X} = a_3,\ a_5, a_1, a_3$	Moment 4;
$\boldsymbol{X} = a_3,\ a_5, a_1, a_3, a_5$	Moment 5;
$\boldsymbol{X} = a_3,\ a_5, a_1, a_3, a_5, a_2$	Moment 6;
$\boldsymbol{X} = a_3,\ a_5, a_1, a_3, a_5, a_2, a_4$	Moment 7;
$\boldsymbol{X} = a_3,\ a_5, a_1, a_3, a_5, a_2, a_4, a_1$	Moment 8;

- This Markov info source has a memory length of 2.

State of Markov Info Source

- The info source randomly sends a symbol at a specific moment. This symbol is only correlated to the m symbols previously transmitted, but it is uncorrelated to other symbols more previously transmitted. The memory length is m.
- The previously transmitted m symbols, which are correlated to the current symbol, are regarded as the state of the Markov info source.

State at moment m: $s_i = \{a_{k_1}, a_{k_2}, \cdots, a_{k_m}\},\ i = 1,2,\cdots,n^m$

$a_{k_1}, a_{k_2}, \cdots, a_{k_m} \in \{a_1,\ a_2, \cdots, a_n\}$

Symbol at moment $(m+1)$: $a_{k_1}, a_{k_2}, \cdots, a_{k_m}, a_{k_{m+1}}$

State at moment $(m+1)$: $s_j = \{a_{k_2}, \cdots, a_{k_m}, a_{k_{m+1}}\},\ j = 1,2,\cdots,n^m$

$a_{k_2}, \cdots, a_{k_m}, a_{k_{m+1}} \in \{a_1,\ a_2, \cdots, a_n\}$

The state of Markov info source transits from state s_i to state s_j .

State Transition of Markov Source

- The state space of a Markov info source $\boldsymbol{X}$ can be expressed as

$$\begin{pmatrix} S \\ P(S) \end{pmatrix} = \begin{Bmatrix} s_1, & s_2, & \cdots, & s_{n^m} \\ p(s_1), & p(s_2), & \cdots, & p(s_{n^m}) \end{Bmatrix}, \sum_{i=1}^{n^m} p(s_i) = 1\,.$$

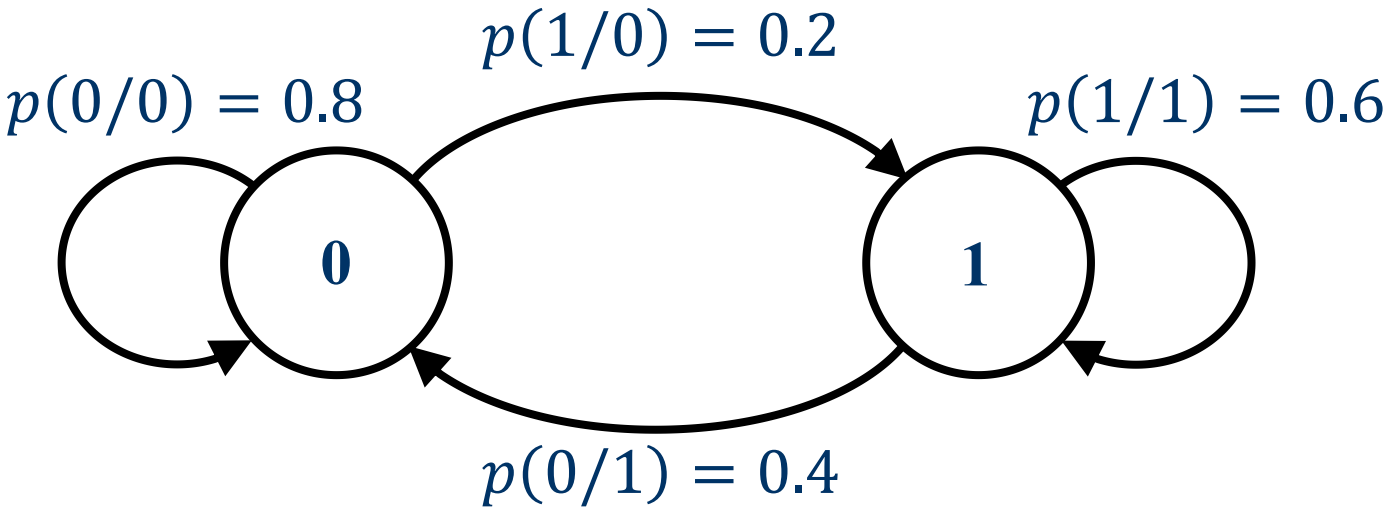
- At the moment $(m+1)$, the Markov info source $\boldsymbol{X}$ transits from state s_i to state s_j.

$$s_i = \{a_{k_1}, a_{k_2}, \cdots, a_{k_m}\} \xrightarrow{p(s_j/s_i)} s_j = \{a_{k_2}, \cdots, a_{k_m}, a_{k_{m+1}}\}$$

- The state transition probability is $p(s_j/s_i)$. The state transition happens since the source sends a symbol $a_{k_{m+1}}$. Therefore, the state transition probability is also the probability of the source sending the symbol $a_{k_{m+1}}$ at the current state s_i, namely $p(s_j/s_i) = p(a_{k_{m+1}}/s_i)$.

State Transition Graph

- All the states having m symbols constitute a stationary Markov chain having finite states. Therefore, a discrete info source having a memory length of m is a Markov info source having an order of m.
- We have a binary Markov info source having an order of one. Its symbol sending probabilities are given by $p(0/0) = 0.8$, $p(1/0) = 0.2$, $p(0/1) = 0.4$, $p(1/1) = 0.6$. Please draw the state transition graph for this Markov info source.



State Transition Graph

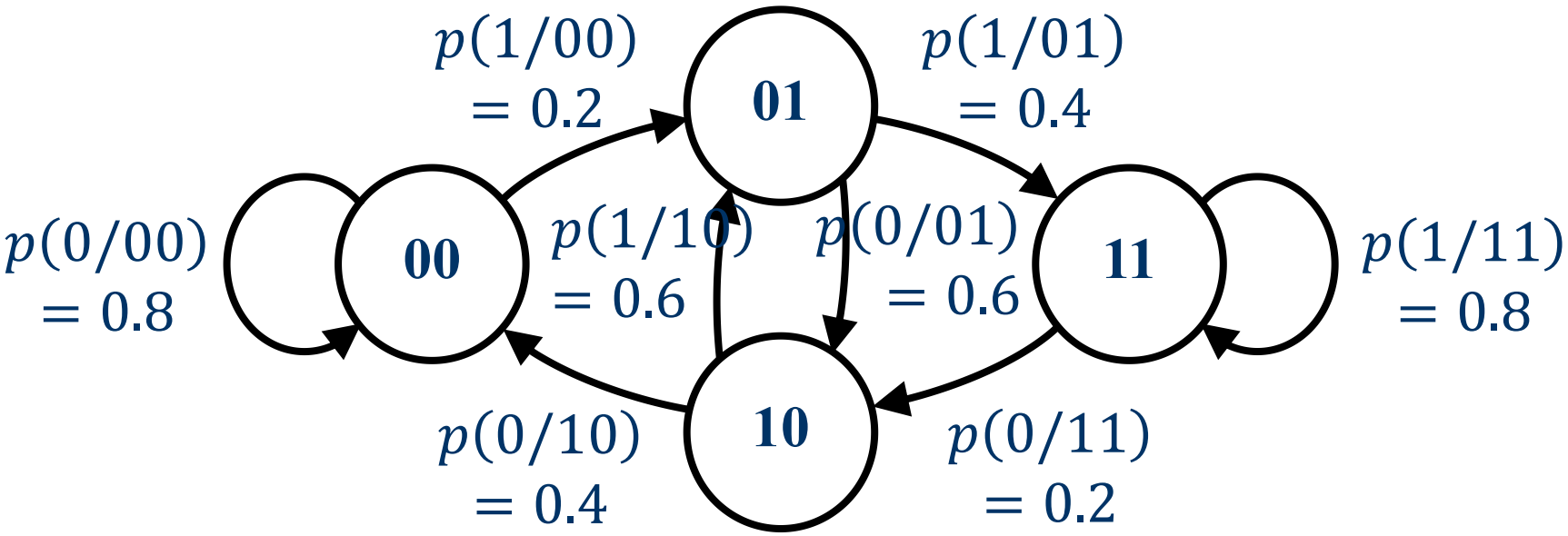
- All the states having m symbols constitute a stationary Markov chain having finite states. Therefore, a discrete info source having a memory length of m is a Markov info source having an order of m.
- We have a binary Markov info source having an order of one. Its symbol sending probabilities are given by $p(0/0) = 0.8$, $p(1/0) = 0.2$, $p(0/1) = 0.4$, $p(1/1) = 0.6$. Please draw the state transition graph for this Markov info source.

State Transition Graph

- We have a binary Markov info source having an order of two. Its symbol sending probabilities are given by $p(0|00) = 0.8, p(1|00) = 0.2$, $p(0|01) = 0.6$, $p(1|01) = 0.4$, $p(0|10) = 0.4$, $p(1|10) = 0.6$, $p(0|11) = 0.2$, $p(1|11) = 0.8$. Please draw the state transition graph for this Markov info source.

State Transition Graph

- We have a binary Markov info source having an order of two. Its symbol sending probabilities are given by $p(0|00) = 0.8, p(1|00) = 0.2$, $p(0|01) = 0.6$, $p(1|01) = 0.4$, $p(0|10) = 0.4$, $p(1|10) = 0.6$, $p(0|11) = 0.2$, $p(1|11) = 0.8$. Please draw the state transition graph for this Markov info source.



Main Content

1. Definition of Markov Source
2. Model of Markov Source
3. Calculation of Entropy Limit
4. Redundancy of Info Source

https://www.bilibili.com/video/BV1Tf4y1n7U6/?spm_id_from=333.788.recommend_more_video.-1

Entropy of Markov Info Source (1)

- For an N-D multi-symbol memory info source, the correlation among the symbols is characterised by joint probabilities. For a Markov info source, the correlation is characterised by state transition probabilities.
- The symbol sent by the Markov info source $\boldsymbol{X}$ having an order of m at the moment $(m+1)$ is correlated to the symbols sent during the previous m moments, which constitute the current state of the source $\boldsymbol{X}$ at the current moment.
- The amount of information carried by a single symbol in average is

$$H_\infty(\boldsymbol{X}) = \lim_{N\to\infty} H(X_N/X_1X_2\cdots X_{N-1})$$

$$= H(X_{m+1}/X_1X_2\cdots X_m) = H_{m+1}(\boldsymbol{X})$$

The entropy rate of the Markov source is equal to the conditional entropy having the same order.

Entropy of Markov Info Source (2)

- The entropy of a Markov source having an order of m can be further expressed as

$$H_\infty(\boldsymbol{X}) = H(X_{m+1}|X_1X_2\cdots X_m)$$

$$= -\sum_{i=1}^{n^m}\sum_{j=1}^{n^m} p(s_i)p(s_j|s_i)\log_2 p(s_j|s_i),$$

where $\{p(s_i)|i = 1,2,\cdots,n^m\}$ are the stationary distribution of the Markov source having an order of m.

Proof: Exploit the definition of the conditional entropy and the mapping relationship between the state transition probabilities and the symbol sending probabilities.

- Exploit the limited memory length of the stationary Markov source to transform the entropy limit to the conditional entropy.

Stationary Distribution

- If a Markov source is time-homogeneous, its stationary distribution $\{p(s_i)|i = 1,2,\cdots,n^m\}$ satisfies:

$$p(s_j) = \sum_{i=1}^{n^m} p(s_i)\, p(s_j/s_i),\ j = 1,2,\cdots,n^m$$

- The matrix form of the equations is formulated as

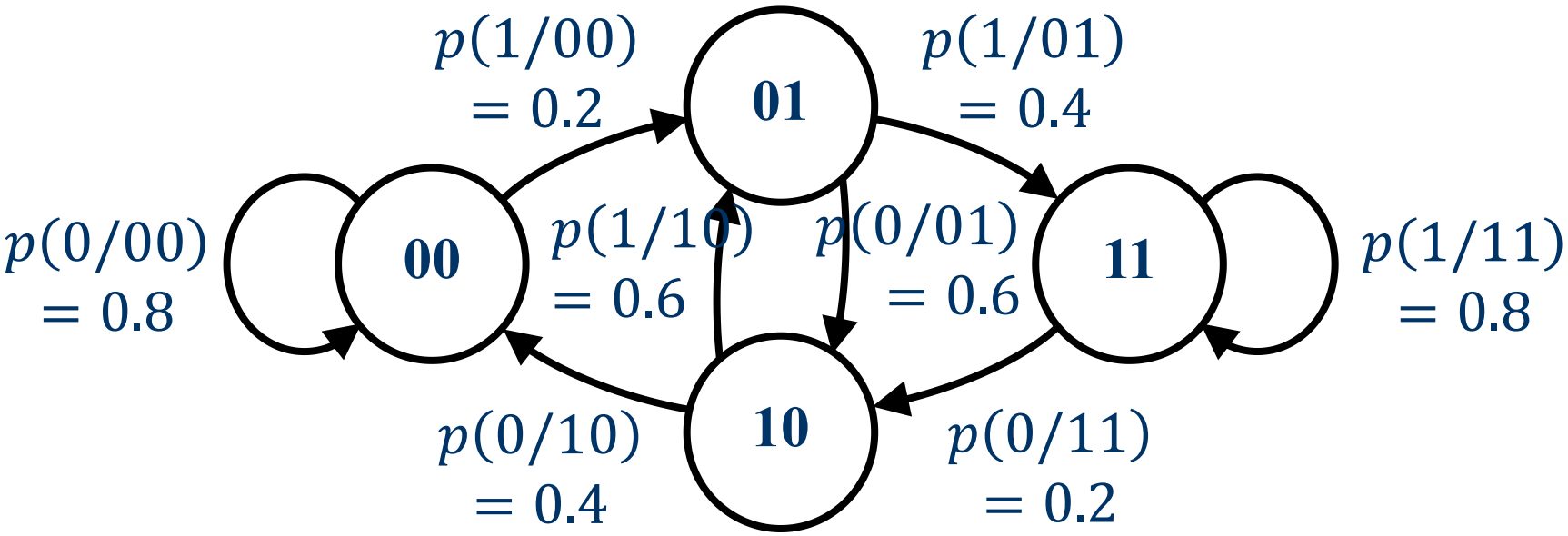
$$\begin{bmatrix} p(s_1/s_1), & p(s_1/s_2), & \cdots, & p(s_1/s_{n^m}), \\ p(s_2/s_1), & p(s_2/s_2), & \cdots, & p(s_2/s_{n^m}), \\ \vdots & \vdots & \ddots & \vdots \\ p(s_{n^m}/s_1), & p(s_{n^m}/s_2), & \cdots, & p(s_{n^m}/s_{n^m}), \end{bmatrix} \times \begin{bmatrix} p(s_1) \\ p(s_2) \\ \vdots \\ p(s_{n^m}) \end{bmatrix} = \begin{bmatrix} p(s_1) \\ p(s_2) \\ \vdots \\ p(s_{n^m}) \end{bmatrix}$$

State Transition Matrix $\boldsymbol{\Lambda}$ Stationary Distribution π

- Additional constraints: $0 \le p(s_j) \le 1,\ \sum_{j=1}^{n^m} p(s_j) = 1$

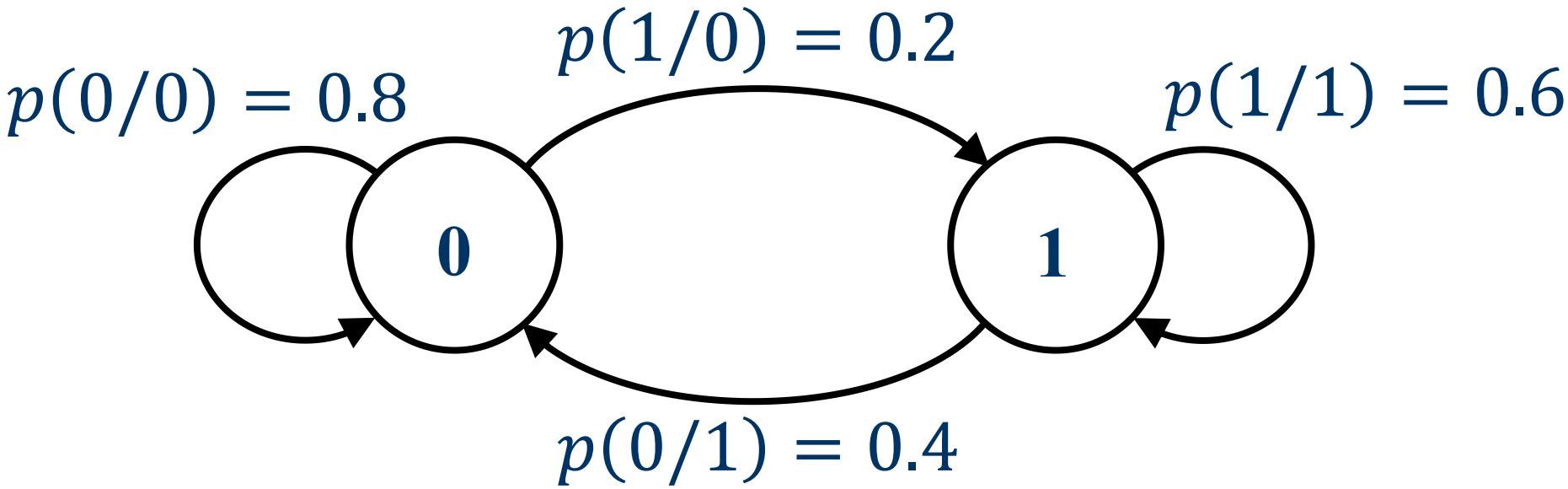
State Transition matrix

- We have a binary Markov info source having an order of two. Its symbol sending probabilities are given by $p(0|00) = 0.8, p(1|00) = 0.2$, $p(0|01) = 0.6$, $p(1|01) = 0.4$, $p(0|10) = 0.4$, $p(1|10) = 0.6$, $p(0|11) = 0.2$, $p(1|11) = 0.8$. Please write the state transition matrix for this Markov info source.



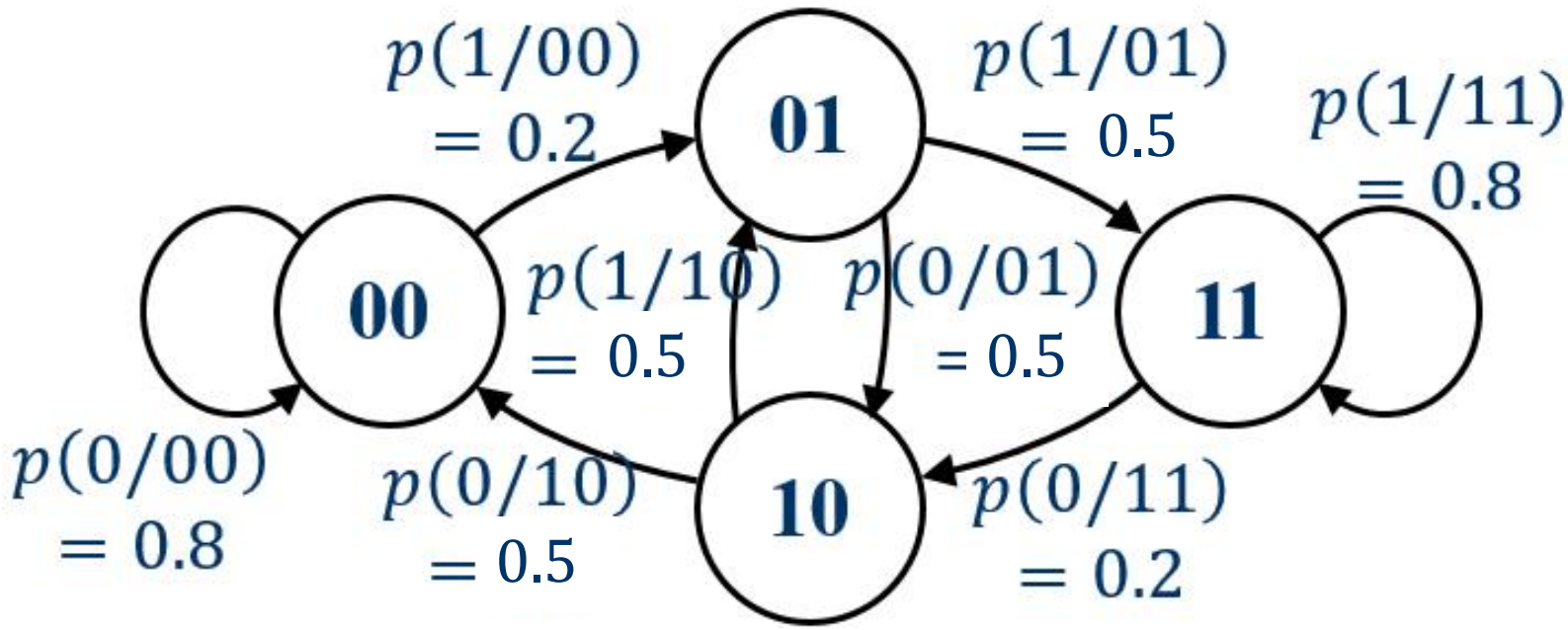
Example

- Please calculate the entropy of the following Markov source having an order of one.



Example

- Please calculate the entropy of the following Markov source having a order of two.



$$H_\infty(\boldsymbol{X}) = H(X_{m+1}|X_1X_2\cdots X_m)$$

$$= -\sum_{i=1}^{n^m}\sum_{j=1}^{n^m} p(s_i)p(s_j|s_i)\log_2 p(s_j|s_i),$$

Solution to Example 2

- Define the four states as E_1, E_2, E_3, E_4,
- Model

$$p(a_j \mid E_i) = \begin{array}{c} \\ E_1(00) \\ E_2(01) \\ E_3(10) \\ E_4(11) \end{array} \begin{array}{c} \begin{array}{cc} a_1(0) & a_2(1) \end{array} \\ \begin{bmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \\ 0.5 & 0.5 \\ 0.2 & 0.8 \end{bmatrix} \end{array}$$

- State Transition Matrix

$$p(E_j \mid E_i) = \begin{array}{c} \\ E_1 \\ E_2 \\ E_3 \\ E_4 \end{array} \begin{array}{c} \begin{array}{cccc} E_1 & E_2 & E_3 & E_4 \end{array} \\ \begin{bmatrix} 0.8 & 0.2 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0.2 & 0.8 \end{bmatrix} \end{array}$$

Solution to Example 2

- stationary distribution can be solved by
  - Define final stationary distribution as Q()

$$\begin{cases} 0.8Q(E_1) & +0.5Q(E_3) & & = Q(E_1) \\ 0.2Q(E_1) & +0.5Q(E_3) & & = Q(E_2) \\ & 0.5Q(E_2) & +0.5Q(E_4) & = Q(E_3) \\ & 0.2Q(E_2) & +0.8Q(E_4) & = Q(E_4) \\ Q(E_1)+Q(E_2)+Q(E_3)+Q(E_4)=1 & & & \end{cases}$$

  - And we get

$$Q(E_1)=\frac{5}{14},\quad Q(E_2)=\frac{1}{7}$$

$$Q(E_3)=\frac{1}{7},\quad Q(E_4)=\frac{5}{14}$$

Solution to Example 2

- Calculate the entropy by using the stationary distribution and the state transition probabilities of the source.

$$H(X \mid E_1) = H(0.8, 0.2) = \text{ 0.722bit/symbol}$$

$$H(X \mid E_2) = H(X \mid E_3) = H(0.5, 0.5) = \text{ 1bit/symbol}$$

$$H(X \mid E_4) = H(0.2, 0.8) = \text{ 0.722bit/symbol}$$

$$H_{m+1} = \sum_i Q(E_i) H(X \mid E_i)$$

$$= \frac{5}{14} \times 0.722 + \frac{1}{7} \times 1 + \frac{1}{7} \times 1 + \frac{5}{14} \times 0.722 = 0.80 bit / \text{ symbol}$$

Steps for Entropy Calculation

STEP 1 Decide the states of the Markov source by considering the legitimate symbol space and the memory length.

STEP 2 Decide the state transition probabilities by considering the conditional sending probabilities of the symbols.

STEP 3 Obtain the stationary distribution of the source by using the state space and the state transition probabilities.

STEP 4 Calculate the entropy by using the stationary distribution and the state transition probabilities of the source.

Modelling a Practical Info Source

- Non-stationary info source's entropy rate does not always exist. We may assume a N-D multi-symbol stationary info source, obtain the joint probabilities $p(X_1 \cdots X_N)$ when N is sufficiently large, calculate the average entropy $H_N(\boldsymbol{X})/N$ for approximating the entropy rate.
- We may also assume a Markov source of order m. The entropy rate is thus its conditional entropy of order m. The practical source may be regarded as a Markov source having $m \to \infty$.

Memoryless Equal Prob. Source $\longrightarrow$ Memoryless Source $\longrightarrow$ Markov Source of Order 1

$H_0 = \log_2 n \quad \geq \quad H_1 \quad \geq \quad H_2$

$\downarrow$

Practical Source $\longleftarrow \cdots \longleftarrow$ Markov Source of Order m $\longleftarrow \cdots$

$H_\infty \quad \leq \quad \leq \quad H_{m+1} \quad \leq$

English Language

Char	Prob	Char	Prob	Char	Prob	Char	Prob
␣	0.2	I	0.055	C	0.023	B	0.0105
E	0.105	R	0.054	F,U	0.0225	V	0.008
T	0.072	S	0.052	M	0.021	K	0.003
O	0.0654	H	0.047	P	0.0175	X	0.002
A	0.063	D	0.035	Y,W	0.012	J,Q	0.001
N	0.059	L	0.029	G	0.011	Z	0.001

- Modelling English with Markov info sources having different orders, we may obtain the approximate entropy of English:

$$H_0 = 4.76\ bit/char \qquad H_1 = 4.03\ bit/char$$

$$H_2 = 3.32\ bit/char \qquad \dots \qquad H_\infty = 1.4\ bit/char$$

Redundancy of Info Source

- A practical info source only needs to transmit the information amount H_∞. It is impossible to obtain an accurate H_∞. We can only approximate it by using Markov info source of order m. Since we have $H_\infty < H_{m+1}$, if we adopt a transmission scheme for transmitting the information amount H_{m+1}, it wastes the precious communication resources and it hence reduces the efficiency.
- Entropy efficiency is defined as the ratio between the actual rate of the practical source and the maximum achievable entropy $\eta = H_\infty / H_0$.
- The redundancy of the information source is thus defined as

$$\xi = 1 - \eta = \frac{H_0 - H_\infty}{H_0}.$$

- Reliability: high redundancy may counteract the adverse effect of the imperfect information transmission.

Meaning of Redundancy

- Source redundancy can well reflect the strength of the dependence between symbols in the sequence of symbols output by discrete sources.
- The larger the redundancy, the smaller the actual entropy of the source. This indicates that the stronger the dependence between source symbols, that is, the longer the memory length between symbols.
- On the contrary, the smaller the redundancy is, which indicates that the weaker the dependence between source symbols is, that is, the shorter the memory length between symbols is.

English Language

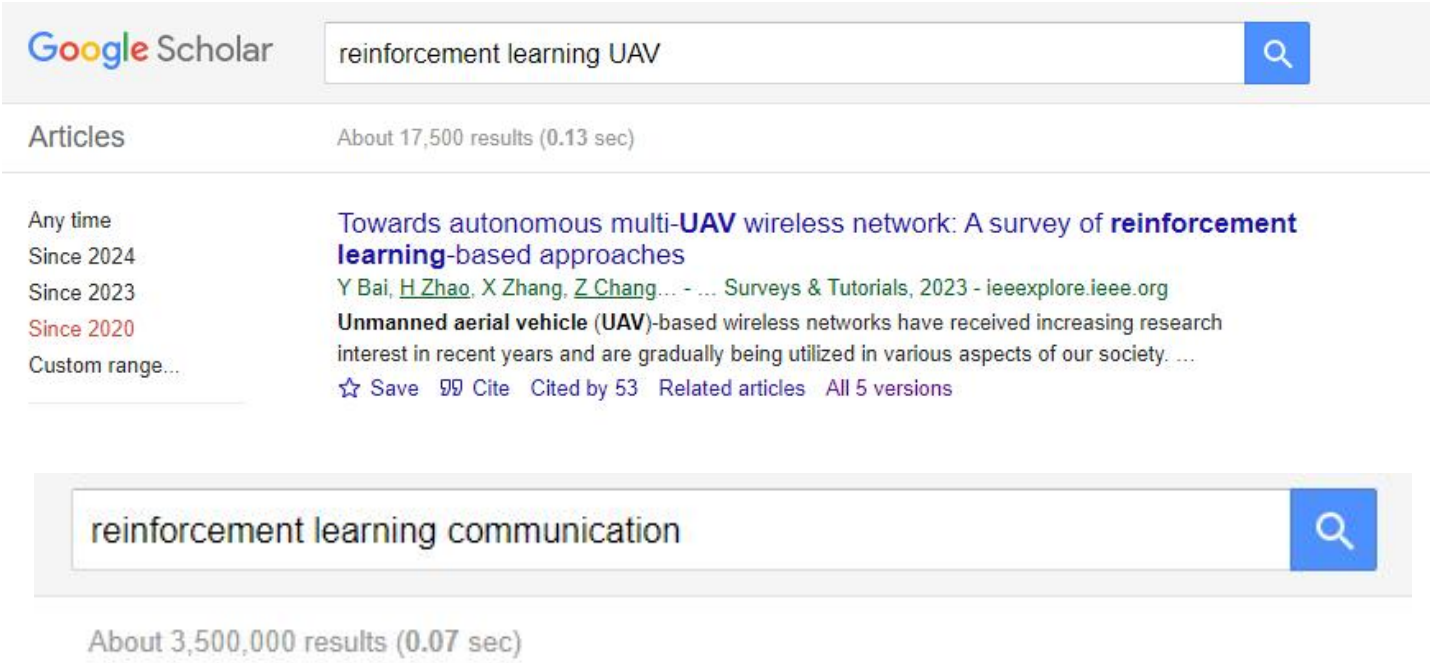
Char	Prob	Char	Prob	Char	Prob	Char	Prob
␣	0.2	I	0.055	C	0.023	B	0.0105
E	0.105	R	0.054	F,U	0.0225	V	0.008
T	0.072	S	0.052	M	0.021	K	0.003
O	0.0654	H	0.047	P	0.0175	X	0.002
A	0.063	D	0.035	Y,W	0.012	J,Q	0.001
N	0.059	L	0.029	G	0.011	Z	0.001

$H_0 = 4.76\ bit/char \qquad H_1 = 4.03\ bit/char$

$H_2 = 3.32\ bit/char \qquad \dots \qquad H_\infty = 1.4\ bit/char$

$$\xi = 1 - \eta = \frac{H_0 - H_\infty}{H_0} = 0.71.$$

MDP & RL



https://www.bilibili.com/video/BV1o14y1a7FL?p=1&vd_source=508ff05d78f989853fcf5d0c8a0d822b

David Silver

https://www.davidsilver.uk/

- David Silver is a principal research scientist at DeepMind and a professor at University College London.
- David’s work focuses on artificially intelligent agents based on reinforcement learning. David co-led the project that combined deep learning and reinforcement learning to play Atari games directly from pixels (Nature 2015).
- He also led the AlphaGo project, culminating in the first program to defeat a top professional player in the full-size game of Go (Nature 2016), and the AlphaZero project, which learned by itself to defeat the world’s strongest chess, shogi and Go programs (Nature 2017, Science 2018). Most recently he co-led the AlphaStar project, which led to the world’s first grandmaster level StarCraft player (Nature 2019).
- His work has been recognised by the Marvin Minsky award, M